

CSCI 264 — Assembly Language

Lab 5

Building an Adder/Subtractor

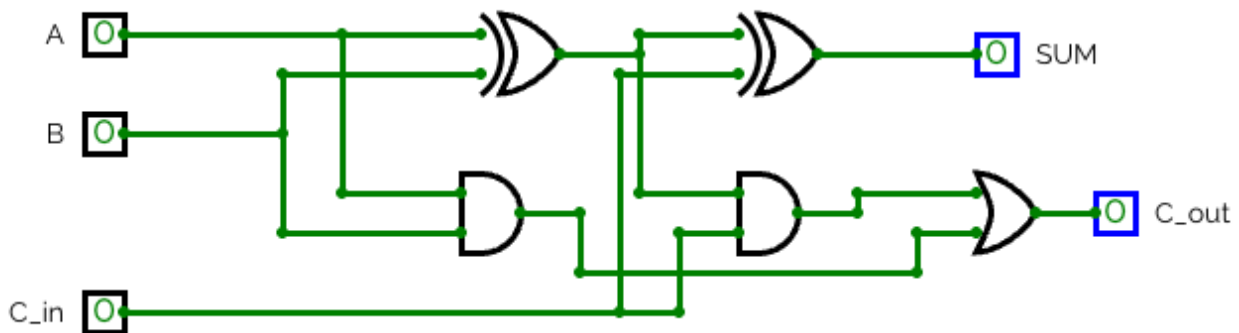
Getting Started

Open <https://circuitverse.org/simulator> in a browser. You do not need an account. The panel on the left holds the circuit elements you need: **Input** and **Output** under *Inputs* and *Outputs*, and **AND**, **OR**, **NOT**, and **XOR** under *Gates*. Click an element in the panel to add it to the simulation. Click an input in the simulation to toggle it between 0 and 1.

Export your circuit as an image as soon as Part 1 works, and again after each part. If the tab closes, you lose all your work.

Part 1 — 1-bit Full Adder

In the top left (next to where it says Main), click the + to create a new circuit tab and name it FA. Build the full adder that we did in class: two half adders and an OR gate. Each half adder is one XOR gate (the sum) and one AND gate (the carry).



Test it against the truth table for at least these four rows:

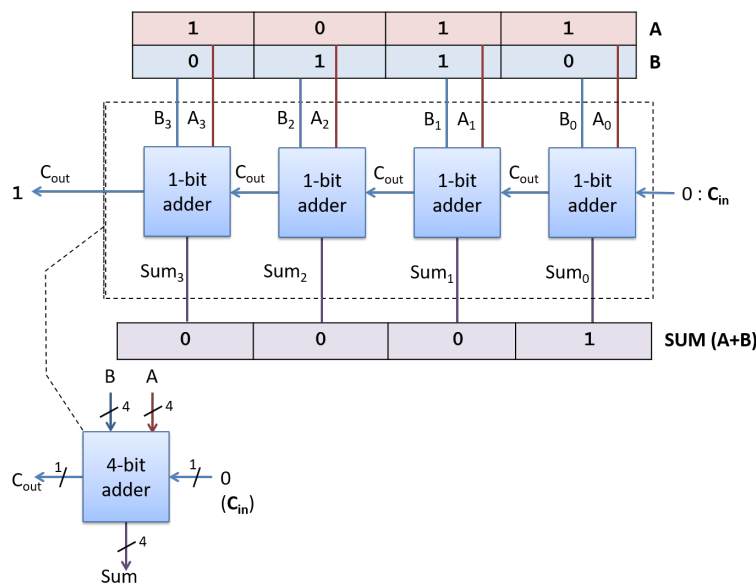
A	B	Cin	SUM	Cout	
0	0	0	0	0	
0	1	0	1	0	
1	1	0	0	1	carry out, no carry in
1	1	1	1	1	1 + 1 + 1 = 3, which is binary 11

Before you move on: deselect any element, so you see the 'Project Properties' panel on the right side, and click 'Edit Layout'. Swap the inputs and outputs in the layout so the inputs are on the right side of the box and the outputs on the left.

Part 2 — 4-bit Ripple Adder

Switch to your main circuit tab, in the menus at the top select **Circuit, Insert Subcircuit** and add four copies of the full adder. Wire them together into a ripple adder:

- Inputs $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$, one bit to each adder.
- The C_{out} of each adder feeds the C_{in} of the adder to its *left* (the next higher bit).
- The C_{in} of the rightmost adder (A_0, B_0) is 0 for now.
- Outputs $S_3S_2S_1S_0$, plus the C_{out} of the leftmost adder.



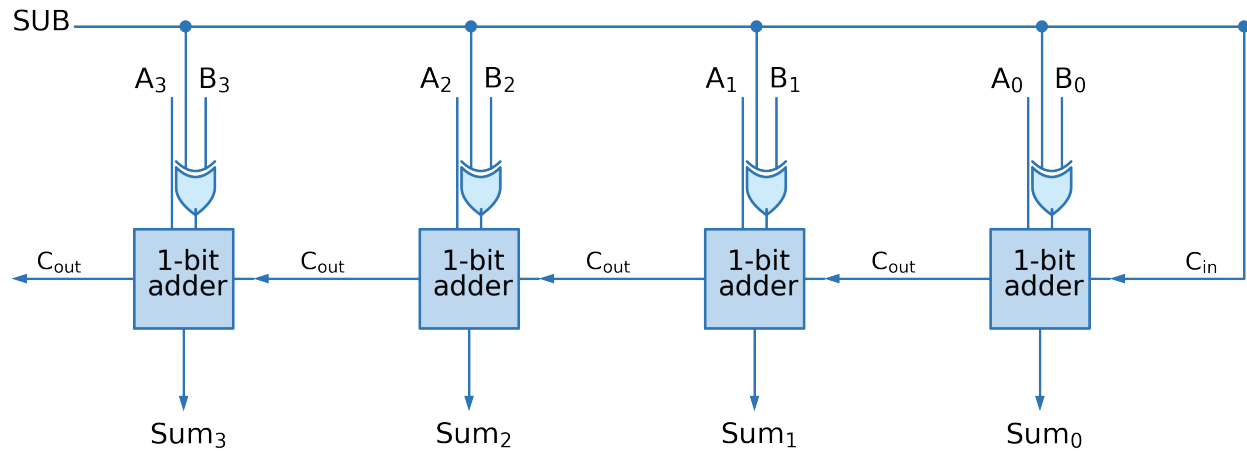
Keep the bits in order on the canvas, with bit 0 on the right, the way you would write the number. Test it with $3 + 4$ and $6 + 2$.

Part 3 — Subtracting with the adder

Add one more input and call it SUB. Then:

1. Put an XOR gate on each B input. One input of each XOR is B_i ; the other is SUB. Modify the adder so that the XOR output, not B_i , goes into the adder.

2. Connect SUB to the Cin of the rightmost adder, replacing the 0.



When SUB is 0 the circuit still computes $A + B$. When SUB is 1 it computes $A - B$.

Test it with $6 - 2$ and $6 + 2$.

If You Finish Early

Add an overflow output: one XOR gate whose inputs are the carry *out* of the leftmost adder and the carry *into* the leftmost adder. Compare this with the signed overflow rule from Lecture 4.

Submission

Before you leave lab, submit on Canvas: An image of your finished circuit, exported from CircuitVerse, showing the the 4-bit adder/subtractor.

There is no demo for this lab, if you don't finish just submit what you have. Spend your out of class time this week preparing for the exam which will be during next lab period.